Together: Backend LIbrary Application API Endpoints

Overview:

1. ***Book Management API***
    a. *GET /api/books - Retrieves all books.*
    b. *GET /api/books/{id} - Retrieves a specific book by its GUID.*
    c. *POST /api/books - Creates a new book record.*
    d. *PUT /api/books/{id} - Updates an existing book.*
    e. *DELETE /api/books/{id} - Deletes a book from the catalog.*
2. ***Member Management API***
    a. *GET /api/members - Retrieves all registered members.*
    b. *GET /api/members/{id} - Retrieves a specific member by their GUID.*
    c. *POST /api/members - Registers a new member.*
    d. *PUT /api/members/{id} - Updates member information.*
    e. *DELETE /api/members/{id} - Removes a member from the system.*
3. ***Borrowing Management API***
    a. *GET /api/borrowing - Retrieves all active and past borrow records.*
    b. *GET /api/borrowing/history/{memberId} - Retrieves the borrowing history for a specific member.*
    c. *POST /api/borrowing/borrow - Processes a book checkout request.*
    d. *POST /api/borrowing/return/{id} - Processes a book return request.*

1. Books

Base route: `/api/books`

Method	Endpoint	Request Body	Status	Description
GET	/api/books	None	**200**	Returns all books. Results are cached for 2 minutes.
GET	/api/books/{id}	None	**200**	Returns a single book by GUID. Cached for 2 minutes. Returns 404 if not found.
POST	/api/books	CreateBookRequest	**201**	Creates a new book. ISBN must be unique. TotalCopies must be > 0.

				AvailableCopies is automatically set to TotalCopies.
PUT	/api/books/{id}	UpdateBookRequest	200	Updates a book. ISBN must remain unique. AvailableCopies cannot exceed TotalCopies. Returns 404 if not found.
DELETE	/api/books/{id}	None	204	Deletes a book by GUID. Returns 404 if not found.

1.1 CreateBookRequest

Field	Type	Validation
Title	string	Required. Cannot be empty.
Author	string	Required. Cannot be empty.
ISBN	string	Required. Must be unique across all books.
TotalCopies	int	Required. Must be greater than 0.

1.2 UpdateBookRequest

Field	Type	Validation
Title	string	Required. Cannot be empty.
Author	string	Required. Cannot be empty.
ISBN	string	Required. Must be unique, excluding the book being updated.
TotalCopies	int	Required. Must be greater than 0.
AvailableCopies	int	Required. Must be >= 0 and <= TotalCopies.

1.3 BookResponse

Field	Type	Validation
Id	Guid	Unique identifier of the book.
Title	string	Title of the book.
Author	string	Author of the book.
ISBN	string	ISBN of the book.

TotalCopies	int	Total copies owned.
AvailableCopies	int	Copies currently available to borrow.

2. Members

Base route: `/api/members`

Method	Endpoint	Request Body	Status	Description
GET	/api/members	None	200	Returns all registered members.
GET	/api/members/{id}	None	200	Returns a single member by GUID. Returns 404 if not found.
POST	/api/members	CreateMemberRequest	201	Registers a new member. Email must be valid and unique.
PUT	/api/members/{id}	UpdateMemberRequest	200	Updates member details. Email must remain valid and unique. Returns 404 if not found.
DELETE	/api/members/{id}	None	204	Deletes a member by GUID. Returns 404 if not found.

2.1 CreateMemberRequest

Field	Type	Validation
FullName	string	Required. Cannot be empty.
Email	string	Required. Must be a valid email format. Must be unique.

2.2 UpdateMemberRequest

Field	Type	Validation
FullName	string	Required. Cannot be empty.
Email	string	Required. Must be valid. Must be unique, excluding this member.

2.3 MemberResponse

Field	Type	Validation
Id	Guid	Unique identifier of the member.
FullName	string	Full name of the member.
Email	string	Email address of the member.
MembershipDate	DateTime	Date the member was registered (UTC).

3. Borrowing

Base route: `/api/borrowing`

Method	Endpoint	Request Body	Status	Description
GET	/api/borrowing	None	200	Returns all borrow records with associated book and member details.
GET	/api/borrowing/{id}	None	200	Returns a single borrow record by GUID. Returns 404 if not found.
GET	/api/borrowing/history/{memberId}	None	200	Returns all borrow records for a member. Returns 404 if member not found.
POST	/api/borrowing/borrow	CreateBorrowRequest	201	Borrows a book. Fails (409) if no copies available or member already has an active borrow for this book.
POST	/api/borrowing/return/{id}	None	200	Returns a borrowed book. Fails (404) if record not found, fails (409) if already returned.

3.1 CreateBorrowRequest

Field	Type	Validation
BookId	Guid	Required. Must reference an existing book.
MemberId	Guid	Required. Must reference an existing member.

3.2 BorrowRecordResponse

Field	Type	Validation

Id	Guid	Unique identifier of the borrow record.
BookId	Guid	ID of the borrowed book.
MemberId	Guid	ID of the borrowing member.
BookTitle	string	Title of the borrowed book.
MemberFullName	string	Full name of the borrowing member.
BorrowDate	DateTime	Date and time the book was borrowed (UTC).
ReturnDate	DateTime?	Date and time returned. Null if still borrowed.
Status	string	'Borrowed' or 'Returned'.

4. Error Response Format

All errors from every endpoint return the following consistent JSON format:

```
{ "error": "<descriptive message>" }
```

Status	Meaning
200	OK — Request succeeded.
201	Created — Resource was created successfully.
204	No Content — Delete succeeded. No response body.
400	Bad Request — Input validation failed or business rule violated.
404	Not Found — The requested resource does not exist.
409	Conflict — Duplicate email/ISBN, no copies available, or borrow record already returned.
500	Internal Server Error — Unexpected server-side error. Details are logged, not exposed.